Difference Equations

FD into PDE ⇒ (algebraic) difference equation

Example:

temperature — const. thermal diffusivity

$$\boxed{\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}}$$

(1D HEAT EQN.)

1st-order fwd. FD in t (marching variable):

time index

$$\left(\frac{\partial T}{\partial t}\right)_i^n = \frac{T_i^{n+1} - T_i^n}{\Delta t} + \mathcal{O}(\Delta t)$$

x-index

2nd-order central 2nd FD in x:

$$\left(\frac{\partial^2 T}{\partial x^2}\right)_i^n = \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2} + \mathcal{O}(\Delta x^2)$$

FD's into PDE:

$$\overbrace{\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}}^{\text{PDE}}$$

$$\underbrace{\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \frac{T_{i+1}^n - 2T_i^n \quad T_{i-1}^n}{\Delta x^2}}_{\text{difference eqn.}} \; \underbrace{+ \mathcal{O}(\Delta t, \Delta x^2)}_{TE}$$

- observe that as $\Delta x \to 0$, $\Delta t \to 0 \Rightarrow TE \to 0$

$\Rightarrow$ difference eqn. (FDE) is <u>consistent</u>
(we recover original PDE)

- we say FDE "first-order accurate in time and second-order accurate in space"

$$\boxed{\text{Explicit time marching}}$$

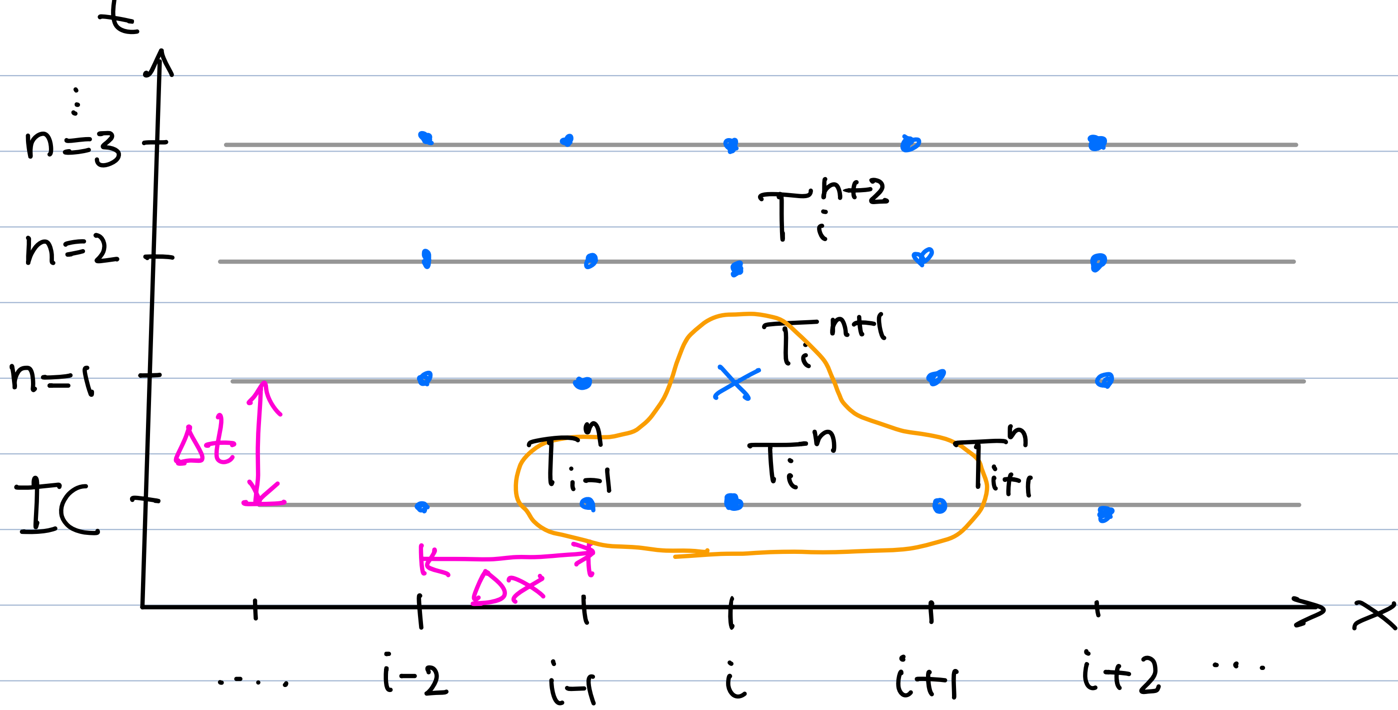
solve for T_i^{n+1} $\Rightarrow$

$$\boxed{\underbrace{T_i^{n+1}}_{\text{unknown}} = \underbrace{T_i^n + \alpha \frac{\Delta t}{\Delta x^2}\left(T_{i+1}^n - 2T_i^n + T_{i-1}^n\right)}_{\substack{\text{known from time level } n \\ (n=1 \,\to\, IC)}}}$$

const.

EULER EXPLICIT

- start computation with IC : $T_i^{n=1} = T_i^{IC}$



stability requirement : $\boxed{\dfrac{\alpha \Delta t}{\Delta x^2} \le \dfrac{1}{2}}$

2-D Heat conduction
(= 2D diffusion eqn.)

PDE

temperature　　const. heat diffusivity

$$\boxed{\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right)}$$

2D HEAT EQN.

Difference eqn. (Euler explicit)

$$T_{i,j}^{n+1} = T_{i,j}^{n} + \alpha \frac{\Delta t}{\Delta x^2}\left(T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}\right)$$

x-index　y-index

$$+ \alpha \frac{\Delta t}{\Delta y^2}\left(T_{i,j+1} - 2T_{i,j} + T_{i,j-1}\right)$$

stability requirement $\boxed{\frac{\alpha \Delta t}{\Delta x^2} + \frac{\alpha \Delta t}{\Delta y^2} \leq \frac{1}{2}}$

accuracy : $O(\Delta t, \Delta x^2, \Delta y^2)$

1-D advection

concentration   bulk velocity

PDE $\boxed{\frac{\partial c}{\partial t} = -u_0 \frac{\partial c}{\partial x}}$

1-D LINEAR ADVECTION

Difference eqn.

- Euler explicit (fwd. first-order in time)
- second-order central FD in x

$\Rightarrow O(\Delta t, \Delta x^2)$

unknown

$$\frac{c_i^{n+1} - c_i^n}{\Delta t} = -u_0 \frac{c_{i+1}^n - c_{i-1}^n}{2\Delta x}$$

$$\Rightarrow \boxed{c_i^{n+1} = c_i^n - \frac{u_0 \Delta t}{2\Delta x}\left(c_{i+1}^n - c_{i-1}^n\right)}$$

stability requirement: UNCONDITIONALLY UNSTABLE!

$$\boxed{\text{MacCormack's method}} \ (1969)$$

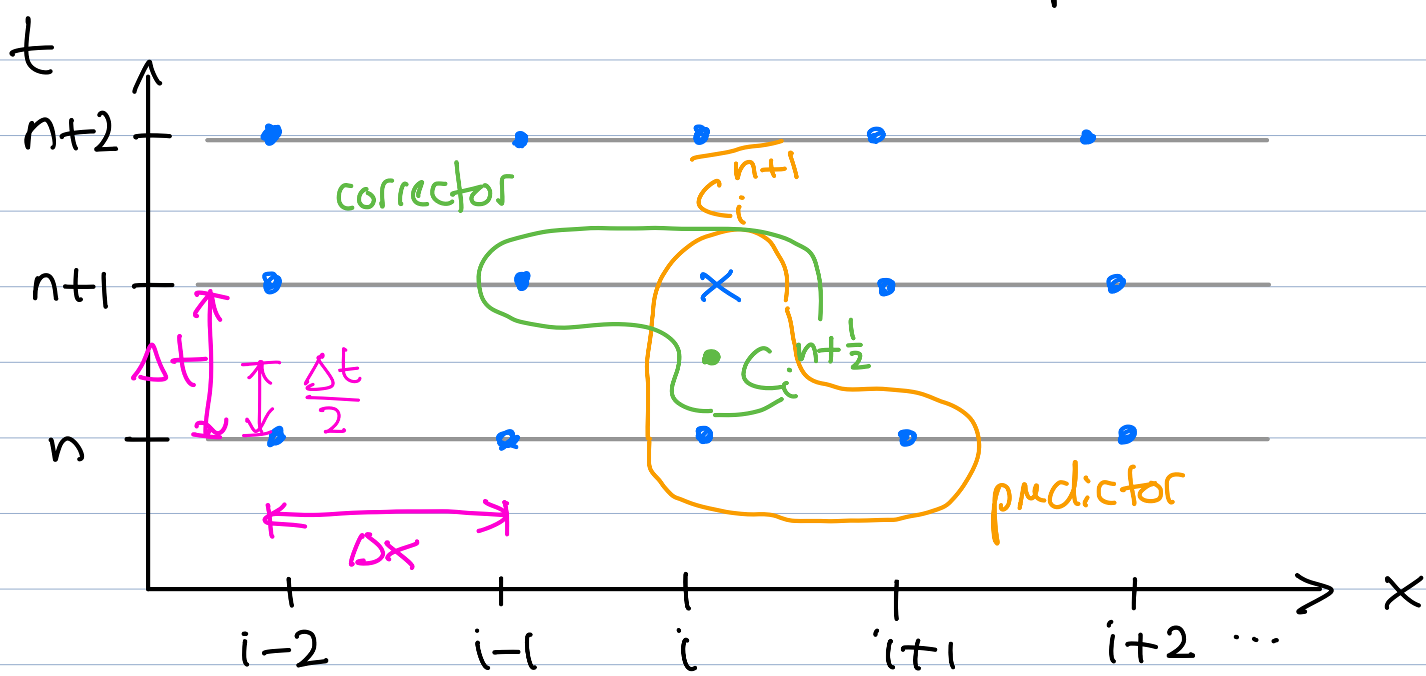
(explicit FD, $\mathcal{O}(\Delta t^2, \Delta x^2)$)

Example:

$$\boxed{\frac{\partial c}{\partial t} + u_0 \frac{\partial c}{\partial x} = 0}$$

1D ADVECTION EQN.

: use first-order FD's for t & x, but introduce an intermediate time step!



- Step 1: forward FD's in t & x (explicit Euler)

$$\frac{\overline{c_i^{n+1}} - c_i^n}{\quad} = -u_0 \frac{c_{i+1}^n - c_i^n}{\quad}$$

$$\frac{\quad}{\Delta t} = \quad u_0 \frac{\quad}{\Delta x}$$

use as temporary, or 'predicted' soln.

- **Step 2**: forward in t from $n+\frac{1}{2}$ to n over $\frac{\Delta t}{2}$ and backward in x

$$\frac{c_i^{n+1} - c_i^{n+\frac{1}{2}}}{\frac{1}{2}\Delta t} = -u_0 \frac{\overline{c_i^{n+1}} - \overline{c_{i-1}^{n+1}}}{\Delta x} \qquad \text{C1}$$

soln C1 for c_i^{n+1}

$$\Rightarrow \quad c_i^{n+1} = c_i^{n+\frac{1}{2}} - \frac{1}{2}\frac{u_0 \Delta t}{\Delta x}\left(\overline{c_i^{n+1}} - \overline{c_{i-1}^{n+1}}\right)$$

: approximate $n+\frac{1}{2}$ as average

$$c_i^{n+\frac{1}{2}} \approx \frac{1}{2}\left(c_i^n + \overline{c_i^{n+1}}\right) \qquad \text{C2}$$

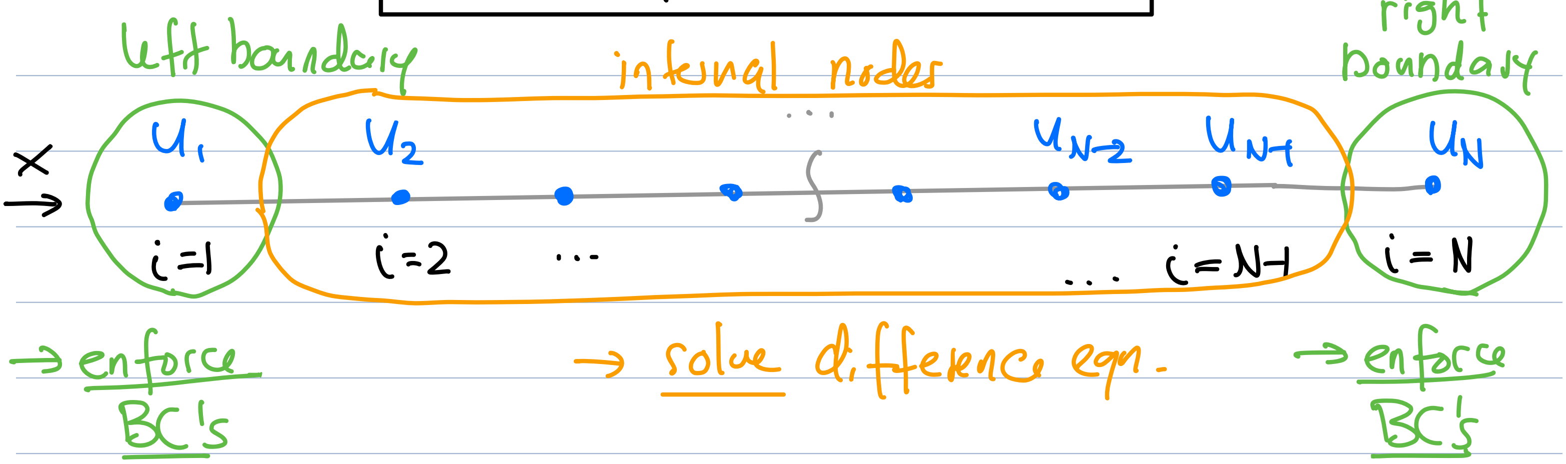
Predictor

$$\boxed{\overline{c_i^{n+1}} = c_i^n - \frac{u_0 \Delta t}{\Delta x}\left(c_{i+1}^n - c_i^n\right)}$$

Corrector

$$\boxed{c_i^{n+1} = \frac{1}{2}\left[\left(c_i^n + \overline{c_i^{n+1}}\right) - \frac{u_0 \Delta t}{\Delta x}\left(\overline{c_i^{n+1}} - \overline{c_{i-1}^{n+1}}\right)\right]}$$

Boundary conditions



• constant value (or Dirichlet) BC

$$u = \text{const.}$$

Ex: i) no-slip condition (zero velocity) at bottom wall

$$u(x, y=0) = 0 \Rightarrow u_{i,j=1} = 0$$

$$(v_{i,j=1} = 0)$$

ii) const. wall temperature

$$T(x, y=0) = T_{wall} \Rightarrow T_{i,j=1} = T_{wall}$$

• constant gradient (or Neumann) BC

$$\frac{\partial u}{\partial x} = \text{const.}$$

Ex: adiabatic wall (no heat flux) at $y=0$

$\rightarrow$ Fourier's Law

heat flux $\left[\frac{W}{m^2}\right]$ thrml. conduct. temperature gradient

$$\dot{q}_y = -k\left(\frac{\partial T}{\partial y}\right)_{y=0} = 0$$

$$\Rightarrow \left(\frac{\partial T}{\partial y}\right)_{y=0} = 0$$

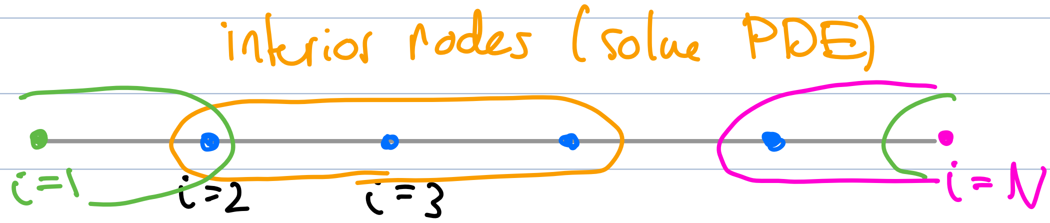
fwd FD $O(\Delta y)$ $\Rightarrow$

$$\frac{T_{i,2} - T_{i,1}}{\Delta y} = 0 \Rightarrow \boxed{T_{i,2} = T_{i,1}}$$

- <u>periodic</u> (not really a BC, but replaces BC)
  $\rightarrow$ get implemented in FD in difference eqn.

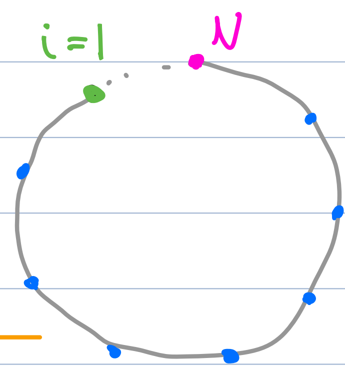
<u>Ex:</u>

$$\frac{\partial^2 T}{\partial x^2} \approx \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta x^2}$$



<u>Ex:</u>

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{2\Delta x}$$



<u>interior</u> :

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{i=3} \approx \frac{u_4 - 2u_3 + u_2}{2\Delta x}$$

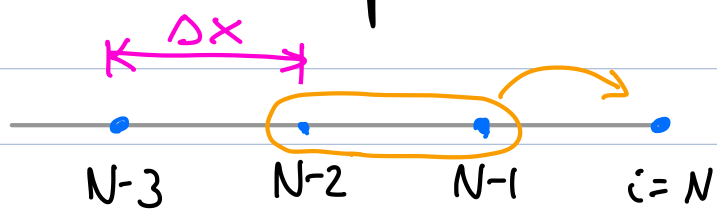
<u>left</u> :

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{i=1} \approx \frac{u_2 - 2u_1 + u_N}{2\Delta x}$$

right: $\left.\dfrac{\partial^2 u}{\partial x^2}\right|_{i=N} \approx \dfrac{u_1 - 2u_N + u_{N-1}}{2\Delta}$

- <u>extrapolation</u> (last resort...)

<u>Ex.</u>: linear extrapolation at right boundary



: construct line from N-2 to N through N-1

$$u_N \approx u_{N-1} + \underbrace{\frac{u_{N-1} - u_{N-2}}{\cancel{\Delta x}}}_{\text{fwd. diff for slope}} \underbrace{2\cancel{\Delta x}}_{\text{distance}}$$

$$\Rightarrow \boxed{u_N \approx -u_{N-2} + 2u_{N-1}} \quad \text{(right)}$$

$$\boxed{u_1 \approx -u_3 + 2u_2} \quad \text{(left)}$$